James M. Rogers

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Education

University of Tennessee, Knoxville December 2025
Bachelor of Science in Physics
Minor in Mathematics
GPA: 3.91/4.0

Technical Skills

Programming languages: CUDA, C, Python, Fortran, McStas neutron ray-tracing
Data Analysis & Tools: SLURM, ROOT, LaTeX, Git/GitHub
Systems & Instrumentation: Beamline optics, PMT construction, Scintillators
Other: Linux administration, NAUI Certified Master Diver / Rescue Diver

Professional Training

LANL Rad Worker II Certification ..... May 2025
LANL Cryogen Worker safety training ..... May 2025
LANL Pressure Safety training ..... May 2025
ORNL Rad Worker I Certification ..... May 2023
NAUI Master Scuba Diver ..... November 2025
NAUI Rescue Scuba Diver ..... May 2025

Honors and Awards

UT-Battelle Scholarship Award ..... June 2022
UT Chancellors Honors Program ..... August 2022 - Present
HOPE Scholarship ..... August 2022 - Present
2 Homer Awards; The Home Depot ..... August 2021 - July 2022
ASUReS Presentation Symposium Winner ..... May 2023
EUReCA Poster Symposium Winner ..... May 2023

Research Experience

Los Alamos National Laboratory (UCNProBe Experiment) Summer 2025
Undergraduate Researcher

- Calibrated detectors and veto system setup for UCNProBe experiment.

- Conducted light-leak testing, beamline component install, and online monitor networking.

University of Tennessee, Department of Physics & Astronomy 2020 – Present
Undergraduate Researcher, Advisor: Prof. Yuri Kamyshkov

- Processed and corrected HFIR GP-SANS data using Fortran, C++, Python, and CUDA.
- Identified background sources in detectors from isotope decay and threshold miscalibration.

Oak Ridge National Laboratory (Neutron Technologies Division) Summer 2023
Undergraduate Researcher, Advisor: Dr. Matthew Frost

- Incorporated reflectivity degradation and misalignment into HFIR CG-2 McStas simulation.
- Reduced simulation intensity discrepancy from $5\times$ to $< 1.2\times$ by parameter optimization.
- First-author paper in *Nuclear Instruments and Methods A* (Nov 2024).

European Spallation Source (NNBAR/HIBEAM Collaboration) 2022 – 2024
Research Contributor

- Designed and simulated a neutron optical focusing device ("Venetian Blinds") in McStas.
- Demonstrated a 40% figure-of-merit improvement over parabolic reflector designs.
- Results presented at the American Conference on Neutron Scattering (2024).
- Contributed to HIBEAM/HighNESS design report and *Journal of Physics G* paper.
- Received two university research excellence awards for related presentations.

Oak Ridge National Laboratory / NREL (Phase-Field Modeling) Summer 2024
Undergraduate Intern

- Implemented and analyzed crystal precipitation models in a high-performance Fortran code.
- Visualized results using ParaView and automated analysis scripts.

Publications

- J. M. Rogers, M. J. Frost, and L. M. Debeer-Schmitt, "Optimization of ray-tracing simulations to confirm performance of the GP-SANS instrument at the High-Flux Isotope Reactor," Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment, vol. 1069, p. 169965, 2024, doi: https://doi.org/10.1016/j.nima.2024.169965.
- V. Santoro et al., "The HIBEAM instrument at the European spallation source," Journal of Physics G: Nuclear and Particle Physics, vol. 52, no. 4, p. 040501, Apr. 2025, doi: 10.1088/1361-6471/adc8c2.
- F. M. Gonzalez et al., "Improved limits on n→ n' transformation from the Spallation Neutron Source," Phys. Rev. D, vol. 110, no. 7, p. 072022, Oct. 2024, doi: 10.1103/PhysRevD.110.072022.
- V. Santoro and others, "HighNESS Conceptual Design Report: Volume I," Sep. 2023, doi: 10.3233/JNR-230950.
- V. Santoro and others, "HighNESS conceptual design report: Volume II. The NNBAR experiment.," J. Neutron Res., vol. 25, no. 3–4, pp. 315–406, 2024, doi: 10.3233/jnr-230951.

Presentations

- *"ESS HIBEAM for Fundamental Neutron Physics."*
  American Conference on Neutron Scattering (Presentation). June 26 2024.
- *"HPC Based Phase-Field Simulations of Liquid Precipitate Boundary Wetting"*
  ORNL PCIP Symposium (Poster). July 29, 2024; ORNL Bldg. 5100.
- *"Novel Neutron Focusing Device for Neutron Oscillation Searches at ESS."*
  ASUReS UTK (Presentation). April 23, 2024. *
- *"Novel Neutron Focusing Device for Neutron Oscillation Searches at ESS."*
  EUReCA UTK (Poster). April 23, 2024. *
- *"Modifying HFIR CG-2 Beamline Simulation to Match Degraded Guide Performance"*
  ORNL URSI/GRSI Symposium (Poster). July 7, 2023; ORNL Bldg. 5100.

*: Received research excellence award from College of Arts and Sciences, UTK.

Outreach & Mentoring

- Founded a student-run "SOUP Club" in high school (2019–present); returned as invited speaker to share research experiences and career advice.
- Mentored new undergraduates in neutron data analysis, preparing documentation and workflows for continuity of UT/ORNL projects.